PROBLEM FORMULATION

Bandit-Based Material Screening Using Epsilon-Greedy Reinforcement Learning

1. Problem Statement

Material screening involves evaluating candidate materials to identify those with promising performance. Testing every candidate equally may use resources on options that perform poorly. This project models screening as a sequential decision-making problem in which an agent chooses one candidate material at a time, observes a simulated performance reward, and uses the feedback to guide later selections. The objective is to maximize cumulative reward over a fixed number of tests.

2. Reinforcement Learning Formulation

Component	Definition
Environment	A simulated material-screening process containing eight candidate materials. Each candidate has a fixed
State / Agent information	The agent maintains an estimated average reward Q(a) and a selection count N(a) for each candidate ma
Action	Select one candidate material (one of eight) for the next simulated test.
Reward	A noisy simulated performance value returned after selecting a material. For candidate a, the reward is s
Policy	Epsilon-Greedy with ε = 0.1. With probability 0.1, select a random candidate (exploration); otherwise, sele
Learning objective	Maximize the cumulative reward obtained over 500 sequential material tests.

3. Reward Model

The simulation assigns the following expected rewards to the eight candidate materials: 2.0, 3.5, 5.0, 4.0, 7.5, 6.0, 3.0 and 8.5. These values are hidden from the learning agent; they are used only by the simulated environment to generate observations. Material 8 has the highest configured expected reward (8.5).

4. Learning Rule

After selecting candidate a and observing reward R, the agent updates its estimated average reward using the sample-average update:

$$Q(a) \leftarrow Q(a) + [R - Q(a)] / N(a)$$

Here, Q(a) is the estimated value of candidate a, N(a) is the number of times it has been selected after the current test, and R is the latest observed reward. Estimates for unselected candidates remain unchanged.

5. Experimental Setup and Evaluation

Parameter	Value
Number of candidate materials	8
Tests per run	500
Independent runs	30
Exploration probability (ε)	0.1

Parameter	Value
Reward noise standard deviation	1.0
Comparison baseline	Uniform random selection

Performance is evaluated by recording cumulative reward over the 500 tests and averaging the learning curves across 30 runs. The Epsilon-Greedy agent is compared with a baseline that selects each material uniformly at random.

6. Assumptions and Scope

This is a simplified educational simulation. Reward distributions are stationary, candidate expected rewards do not change over time, and each test returns one scalar reward. No real laboratory measurements, material synthesis, or physical testing are performed. Therefore, the results demonstrate the bandit decision process only and must not be interpreted as validated scientific findings about real materials.

Deliverable: A Python implementation, learning-curve comparison against random selection, and a short presentation/report.